

WEEK 7 ASSGINMENT

Step 1: Create a New Notebook

Click: + New → **Notebook**

Name it: Delta_Assignment

Language: Python

Step 2: Create the Dataset

Your assignment doesn't provide one, so we'll create it ourselves.

Create a file named **customer_master.csv** with the following data:

CustomerID	Name	City	Age	Email
101	Aman	Delhi	24	aman@gmail.com
102	Priya	Mumbai	27	priya@gmail.com
103	Rahul	Chandigarh	30	rahul@gmail.com
104	Simran	Amritsar	26	simran@gmail.com
105	Karan	Jaipur	29	karan@gmail.com

Open Excel

Create these columns:

CustomerID

Name

City

Age

Email

Fill the data above.

Save As: customer_master.csv

Step 3: Create Incremental Dataset

Create another CSV: **customer_incremental.csv**

CustomerID	Name	City	Age	Email
102	Priya	Pune	28	priya@gmail.com

CustomerID	Name	City	Age	Email
105	Karan	Jaipur	30	karan@gmail.com
106	Neha	Delhi	25	neha@gmail.com
107	Rohit	Lucknow	31	rohit@gmail.com

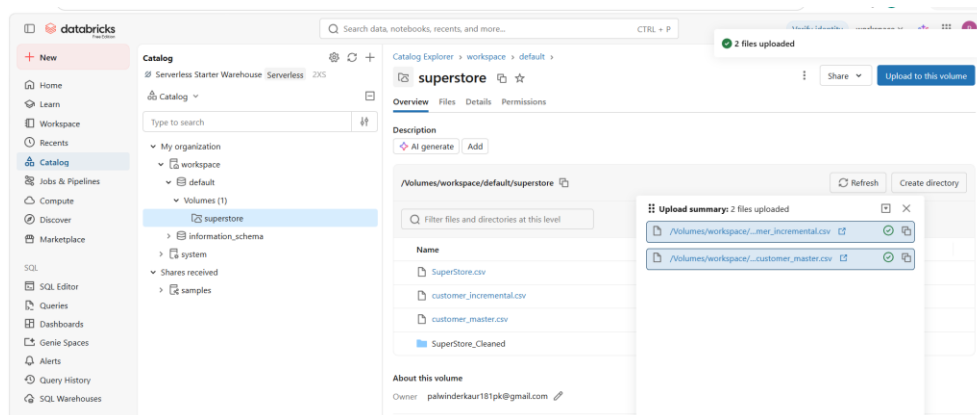
Step 4: Upload Both Files

Go to:

Catalog → workspace → default → Volumes → superstore

Upload:

- customer_master.csv
- customer_incremental.csv



Step 5: Read Master Dataset

Master dataset displayed:

Just now (3s) 1 Python

```

master_df = spark.read \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .csv("/Volumes/workspace/default/superstore/customer_master.csv")

display(master_df)

```

> [See performance \(1\)](#) ✓ Checked for improvements

> master_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]

	CustomerID	Name	City	Age	Email
1	101	Aman	Delhi	24	aman@gmail.com
2	102	Priya	Mumbai	27	priya@gmail.com
3	103	Rahul	Chandigarh	30	rahul@gmail.com
4	104	Simran	Amritsar	26	simran@gmail.co...
5	105	Karan	Jaipur	29	karan@gmail.com

5 rows | 3.264s runtime Refreshed now

Step 6: Handle Missing Values

```
▶ Just now (1s) 2 Python
```

```
master_df = master_df.fillna("Unknown")
```

```
> master_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]
```

Step 7: Remove Duplicates

After cleaning:

```
▶ Just now (<1s) 3
```

```
master_df = master_df.dropDuplicates()
```

```
> master_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]
```

```
▶ Just now (2s) 4 Python
```

```
print("Rows:", master_df.count())
```

```
> See performance (1) Optimize
```

Rows: 5

Step 8: Save as Delta Table

```
▶ 1 minute ago (7s) 11
```

```
master_df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("/Volumes/workspace/default/superstore/customer_delta")
```

```
> See performance (1) ✓ Checked for improvements
```

Step 9: Read Delta Table

Delta table:

```
▶ Just now (3s) 13 Python
```

```
from delta.tables import DeltaTable

delta_df = spark.read.format("delta") \
    .load("/Volumes/workspace/default/superstore/customer_delta")

display(delta_df)
```

```
> See performance (1) ✓ Checked for improvements
```

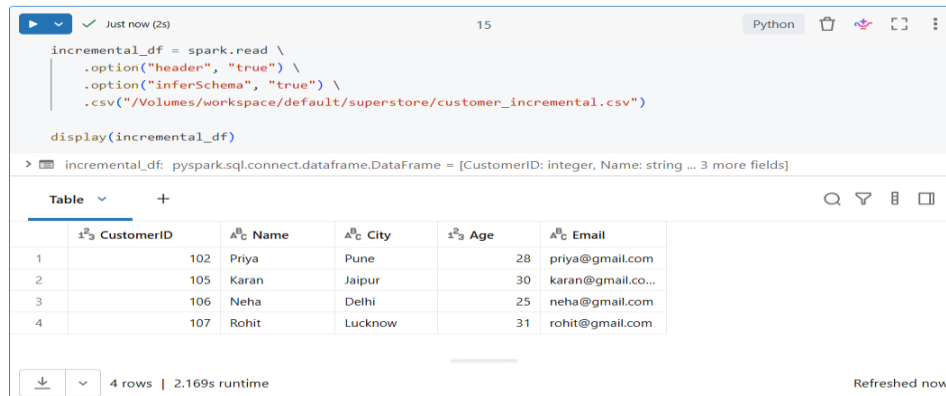
```
> delta_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]
```

	CustomerID	Name	City	Age	Email
1	101	Aman	Delhi	24	aman@gmail.com
2	103	Rahul	Chandigarh	30	rahul@gmail.com
3	104	Simran	Amritsar	26	simran@gmail.co...
4	105	Karan	Jaipur	29	karan@gmail.com
5	102	Priya	Mumbai	27	priya@gmail.com

5 rows | 2.853s runtime Refreshed now

Step 10: Read Incremental Dataset

Incremental dataset:



```
incremental_df = spark.read \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .csv("/Volumes/workspace/default/superstore/customer_incremental.csv")

display(incremental_df)
```

incremental_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]

	CustomerID	Name	City	Age	Email
1	102	Priya	Pune	28	priya@gmail.com
2	105	Karan	Jaipur	30	karan@gmail.co...
3	106	Neha	Delhi	25	neha@gmail.com
4	107	Rohit	Lucknow	31	rohit@gmail.com

4 rows | 2.169s runtime

Refreshed now

Step 11: Perform MERGE

MERGE code executed:



```
from delta.tables import DeltaTable

deltaTable = DeltaTable.forPath(
    spark,
    "/Volumes/workspace/default/superstore/customer_delta"
)

deltaTable.alias("old").merge(
    incremental_df.alias("new"),
    "old.CustomerID = new.CustomerID"
).whenMatchedUpdateAll() \
.whenNotMatchedInsertAll() \
.execute()
```

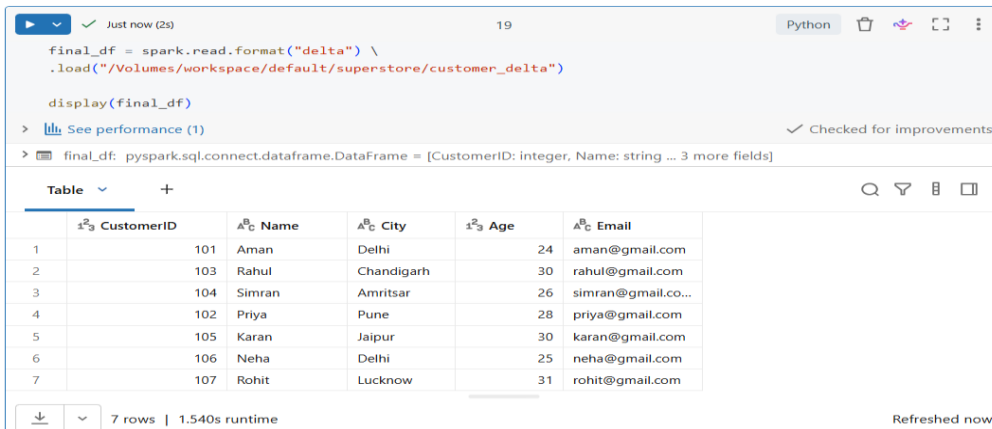
See performance (1)

Optimize

DataFrame[num_affected_rows: bigint, num_updated_rows: bigint, num_deleted_rows: bigint, num_inserted_rows: bigint]

Step 12: Validate

Final merged data:



```
final_df = spark.read.format("delta") \
    .load("/Volumes/workspace/default/superstore/customer_delta")

display(final_df)
```

See performance (1)

Checked for improvements

final_df: pyspark.sql.connect.dataframe.DataFrame = [CustomerID: integer, Name: string ... 3 more fields]

	CustomerID	Name	City	Age	Email
1	101	Aman	Delhi	24	aman@gmail.com
2	103	Rahul	Chandigarh	30	rahul@gmail.com
3	104	Simran	Amritsar	26	simran@gmail.co...
4	102	Priya	Pune	28	priya@gmail.com
5	105	Karan	Jaipur	30	karan@gmail.com
6	106	Neha	Delhi	25	neha@gmail.com
7	107	Rohit	Lucknow	31	rohit@gmail.com

7 rows | 1.540s runtime

Refreshed now

Step 13: Count Rows

▶

▼

✓ Just now (1s)

21

Python









```
print("Total Rows:", final_df.count())
```

> [See performance \(1\)](#)

✓ Checked for improvements

Total Rows: 7

Step 14: Check Duplicates

▶

▼

✓ Just now (1s)

23

Python









```
print("Duplicate Rows:",  
      | final_df.count() - final_df.dropDuplicates().count())
```

> [See performance \(1\)](#)

✓ Checked for improvements

Duplicate Rows: 0

Step 15: Save Final Delta Table as CSV (Optional)

▶

▼

✓ 1 minute ago (2s)

25









```
final_df.write \  
| .mode("overwrite") \  
| .option("header", "true") \  
| .csv("/Volumes/workspace/default/superstore/final_customer_data")
```

> [See performance \(1\)](#)